La Scuola d'Italia / Dr. Huson / IB Math HL
28 May 2026 Markscheme

8.1 Classwork: Rational Functions (HL) — Markscheme

1. $f(x) = \dfrac{x^2 - 3x - 4}{x - 2}, \ \ x \neq 2.$

(a) Vertical asymptote: $x = 2$. **A1**

(b) x-intercepts: $x^2 - 3x - 4 = 0 \Rightarrow (x-4)(x+1) = 0$

$(4, 0)$ and $(-1, 0)$ **A1A1**

y-intercept: $f(0) = \dfrac{-4}{-2} = 2$, so $(0, 2)$. **A1**

Accept $x = 4$, $x = -1$ and $y = 2$ in place of the ordered pairs.

(c) Polynomial division of $x^2 - 3x - 4$ by $x - 2$: **M1**

Quotient $x - 1$, remainder -6. **A1**

Hence $f(x) = x - 1 + \dfrac{-6}{x-2} = x - 1 - \dfrac{6}{x-2}$ **AG**

Oblique asymptote: $y = x - 1$. **A1**

M1 may also be awarded for any other valid method, e.g. writing $x^2 - 3x - 4 = (x-2)(x+c) + r$ and matching coefficients to find $c = -1$, $r = -6$.

(d) *GDC sketch: two branches separated by $x = 2$, each approaching the oblique asymptote $y = x - 1$ at the appropriate end; x-intercepts at -1 and 4, y-intercept at 2.*

Correct overall shape with both asymptotes drawn (dashed). **A1**

All three intercepts labelled correctly on the sketch. **A1**

(e) Sign analysis of $\dfrac{(x-4)(x+1)}{x-2}$ on $(-\infty, -1)$, $(-1, 2)$, $(2, 4)$, $(4, \infty)$: **M1**

$f(x) > 0$ on $-1 < x < 2$ or $x > 4$. **A1**

Accept the interval form $(-1, 2) \cup (4, \infty)$. FT: award A1 if intervals are read correctly from the candidate's sketch in (d).

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2. $f(x) = \dfrac{1}{x^2 - 2x - 8}, \ \ x \neq -2, 4.$

(a) Factor denominator: $x^2 - 2x - 8 = (x-4)(x+2)$. **(M1)**

Vertical asymptotes: $x = -2$ and $x = 4$. **A1A1**

Horizontal asymptote: $y = 0$ (denominator has higher degree than numerator). **A1**

(b) $f(0) = \dfrac{1}{-8} = -\dfrac{1}{8}$. **A1**

(c) The denominator $q(x) = x^2 - 2x - 8$ has minimum at $x = 1$ (vertex of the parabola): $q'(x) = 2x - 2 = 0 \Rightarrow x = 1$. **M1**

So $f(x) = \dfrac{1}{q(x)}$ has a stationary point at $x = 1$, with

$f(1) = \dfrac{1}{1-2-8} = -\dfrac{1}{9}$. Coordinates: $\left(1, -\frac{1}{9}\right)$. **A1A1**

On the middle branch $-2 < x < 4$, $q(x) < 0$ with q achieving its minimum (most negative value) at $x = 1$, so $\dfrac{1}{q(x)}$ achieves its maximum there: local *maximum*. **R1**

Accept differentiating f directly using the quotient or chain rule, $f'(x) = -\dfrac{2x-2}{(x^2-2x-8)^2}$, *and solving* $f'(x) = 0$.

(d) *Sketch shows three branches: left branch ($x < -2$) rising from $y = 0^+$ as $x \to -\infty$ up to $+\infty$ as $x \to -2^-$; middle branch ($-2 < x < 4$) an inverted U through $\left(1, -\frac{1}{9}\right)$, with $f \to -\infty$ at both ends; right branch ($x > 4$) from $+\infty$ at $x = 4^+$ down to $y = 0^+$ as $x \to +\infty$.*

Correct overall shape (three branches; left and right branches above the x-axis, middle branch entirely below). **A1**

Asymptotes (dashed), y-intercept and local max labelled. **A1**

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3. $f(x) = \dfrac{x-2}{x^2-9}$, $x \neq \pm 3$.

(a) Factor: $x^2 - 9 = (x-3)(x+3)$.

Vertical asymptotes: $x = 3$ and $x = -3$. **A1A1**

Horizontal asymptote: $y = 0$. **A1**

(b) x-intercept: $x - 2 = 0 \Rightarrow (2, 0)$. **A1**

y-intercept: $f(0) = \dfrac{-2}{-9} = \dfrac{2}{9}$, so $\left(0, \frac{2}{9}\right)$. **A1**

(c) Critical values for the sign chart: $x = -3$, $x = 2$, $x = 3$. Sign of $\dfrac{(x-2)}{(x-3)(x+3)}$ in each interval: **M1**

interval	$x < -3$	$-3 < x < 2$	$x = 2$	$2 < x < 3$	$x > 3$
sign of $f(x)$	$-$	$+$	0	$-$	$+$

Solution: $-3 < x \leq 2 \quad \text{or} \quad x > 3$. **A1A1**

Accept $(-3, 2] \cup (3, \infty)$. *Note:* $x = \pm 3$ *are excluded (undefined);* $x = 2$ *is included since the inequality is non-strict.*

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4. $g(x) = \dfrac{ax+4}{3-x}$, $x \neq 3$.

(a) Method 1 (swap and equate).

Let $y = \dfrac{ay+4}{3-y}$ (swap x and y): $\quad x(3-y) = ay+4$ **M1**

$3x - 4 = y(a+x) \;\Rightarrow\; g^{-1}(x) = \dfrac{3x-4}{x+a}$ **A1**

Equate $g^{-1}(x) = g(x)$ and cross-multiply: **M1**

$(3x-4)(3-x) = (ax+4)(x+a)$, leading on the constant term to $-12 = 4a \Rightarrow a = -3$. **A1**

Method 2 (Möbius trace condition). For a Möbius function $\dfrac{Ax+B}{Cx+D}$, the condition $g^{-1} = g$ (for all x in the domain) is equivalent to $A + D = 0$. Writing g as $\dfrac{ax+4}{-x+3}$ gives $A = a$, $D = 3$, so $a + 3 = 0 \Rightarrow a = -3$. (*equivalent M1 A1 M1 A1*)

(b) Line of symmetry: $y = x$. **A1**

If $g^{-1} = g$, then (p, q) on the graph of g implies (q, p) is also on the graph (since $g(p) = q$ gives $g^{-1}(q) = p$, and $g^{-1} = g$ means $g(q) = p$). The points (p, q) and (q, p) are reflections of one another across the line $y = x$, hence the graph is symmetric about $y = x$. **R1**